

ETL Process Used in My Project

1. Extraction

Extraction Method: Push Extraction

Extraction Type: Incremental Extraction

Extraction Technique: File Parsing

Description:

In this step, data is extracted from multiple source systems (CRM and ERP) in the form of files. Using the Push Extraction method, the data is sent automatically to the data warehouse. Incremental extraction is used to fetch only the new or updated data. File parsing techniques are applied to read and prepare the data for the next stage.

2. Transformation

In this step, the extracted data is cleaned, standardized, and transformed before loading.

The transformation process includes all the steps shown in the ETL image, such as:

Data Cleansing

- Remove Duplicates
- Data Filtering
- Handling Missing Data
- Handling Invalid Values
- Handling Unwanted Spaces
- Data Type Casting
- Outlier Detection

Data Integration

Derived Columns

Data Normalization & Standardization

Business Rules & Logic

Data Aggregations

Data Enrichment

Description:

This step ensures the data is accurate, consistent, and meaningful. It combines data from different sources, applies business rules, and prepares it for loading into the warehouse.

3. Load

Processing Type: Batch Processing

Load Method: Full Load (Truncate and Insert)

Description:

In the loading step, the transformed data is loaded into the target data warehouse tables. The existing data in the target tables is truncated, and the latest cleaned data is inserted. This ensures that only fresh and accurate data is available for reporting and analytics.